

RAG Knowledge Base Deployment

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1. Course Content

- Master the workflow and methods for locally deploying, debugging, and testing a RAG knowledge base.
- Master the method of extending the RAG knowledge base according to your own task scenarios.

Tip

- The RAG knowledge base helps general-purpose AI large models gain reference knowledge in vertical domains, avoiding hallucinated replies from AI large models and increasing the model's ability to reply with vertical domain knowledge.
- The RAG knowledge base can help the robot quickly extend its generalization capability in different task scenarios.

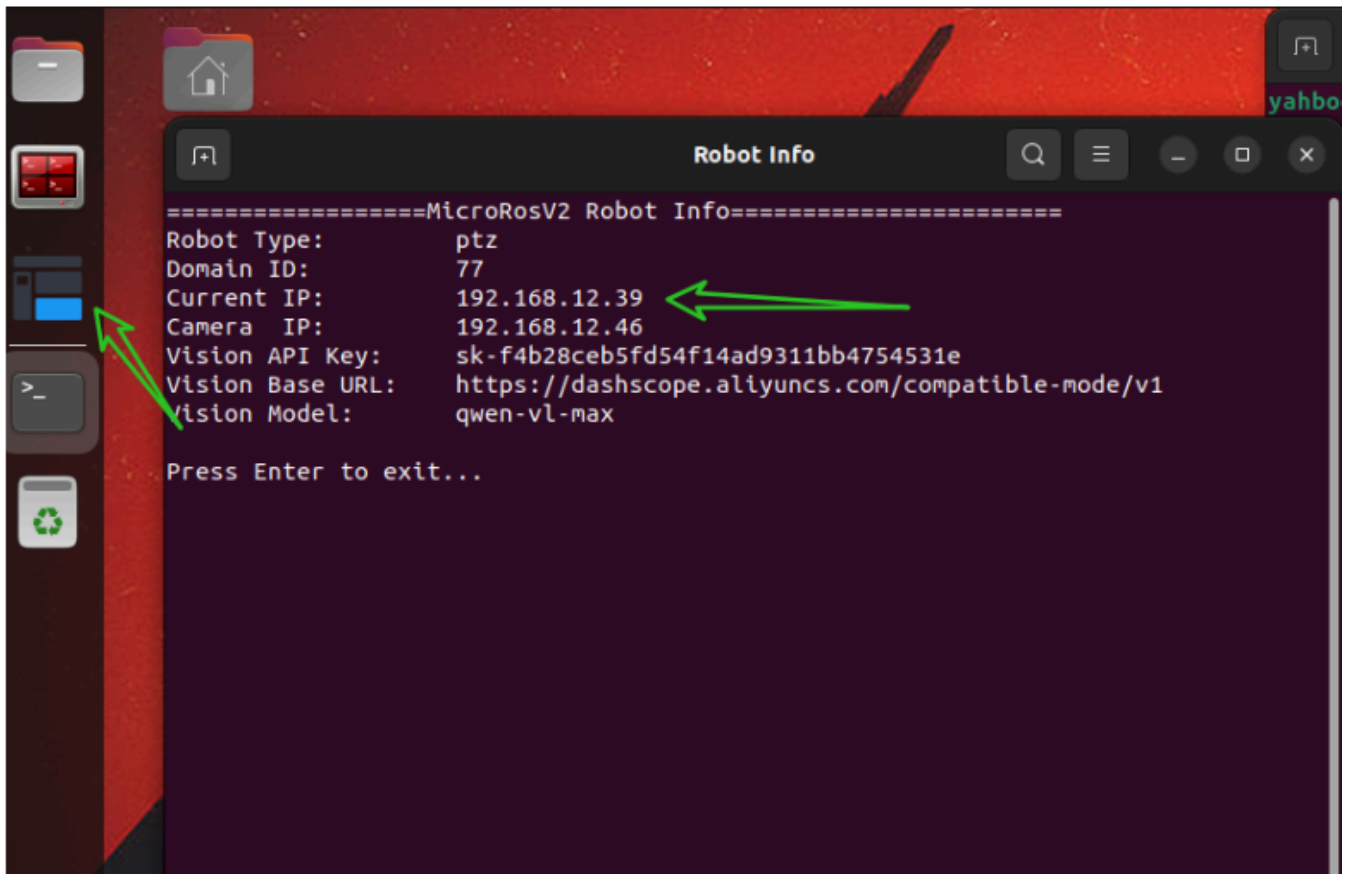
2. Start the Dify Service

- Connect to the vehicle via VNC or SSH, and enter the command in the terminal:

```
sh ~/bringup_dify.sh
```

```
jetson@yahboom: ~  
ROS_DOMAIN_ID: 62 | ROS: humble  
my_robot_type: M1 | my_lidar: c1 | my_camera: usb  
-----  
jetson@yahboom:~$ sh ~/bringup_dify.sh  
WARN[0000] The "DB_USERNAME" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_DATABASE" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_PASSWORD" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_PASSWORD" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_DATABASE" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_USERNAME" variable is not set. Defaulting to a blank string.  
WARN[0000] The "CERTBOT_EMAIL" variable is not set. Defaulting to a blank string  
.  
WARN[0000] The "CERTBOT_DOMAIN" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_DATABASE" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_PASSWORD" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_USERNAME" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_USERNAME" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_DATABASE" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_PASSWORD" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_DATABASE" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_PASSWORD" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_PASSWORD" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_USERNAME" variable is not set. Defaulting to a blank string.  
[+] Running 10/10  
✓ Container docker-sandbox-1      Runni...      0.0s  
✓ Container docker-web-1          Running       0.0s  
✓ Container docker-ssrf_proxy-1   Ru...         0.0s  
✓ Container docker-db-1           Healthy      0.5s  
✓ Container docker-redis-1        Running       0.0s  
✓ Container docker-worker_beat-1   R...         0.0s  
✓ Container docker-plugin_daemon-1 Running       0.0s  
✓ Container docker-api-1          Running       0.0s  
✓ Container docker-worker-1        Runnin...     0.0s  
✓ Container docker-nginx-1        Running       0.0s  
jetson@yahboom:~$
```

- Check the IP address of the virtual machine by clicking "Robot Information" on the left side of the desktop.

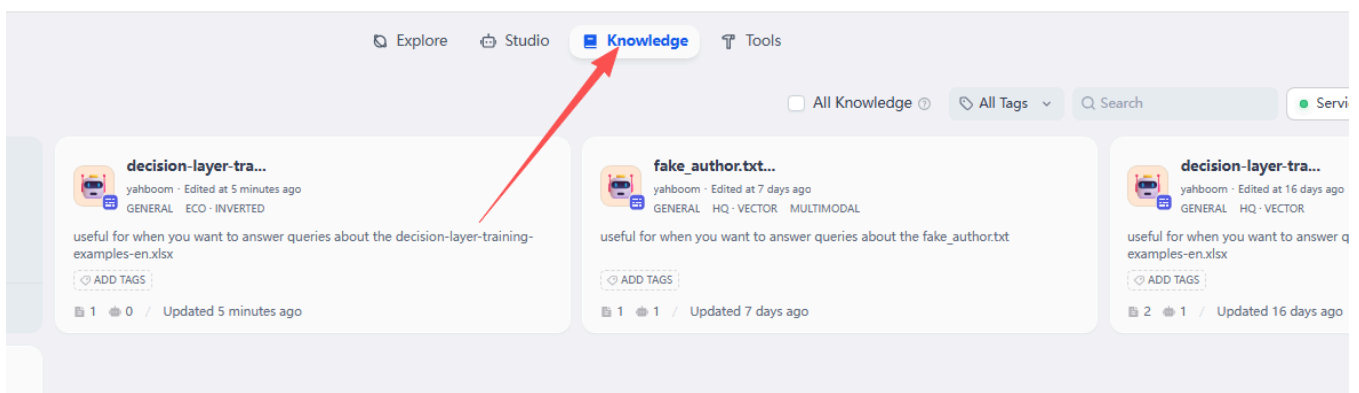


3. View the Preset Knowledge Bases

- Click the "Knowledge" page on the homepage. Dify comes with two preset RAG knowledge bases at the factory, with the same content but different languages.

💡 Tip

- The preset knowledge bases here provide training samples for some task scenarios, helping AI models quickly master the relevant skills.



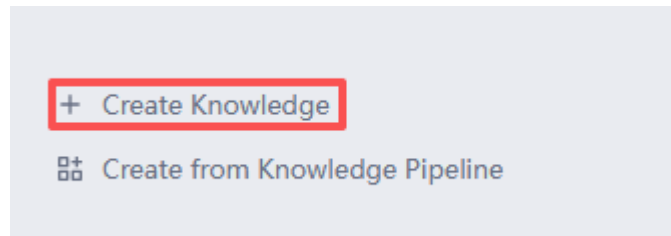
- Open a knowledge base. There are two preset files: `Intent Mapping` and `Decision-Layer Sample Library`, where:
 - Intent mapping: stores user intents and the corresponding tasks the robot needs to execute.
 - Decision-layer sample library: stores some preset reference samples for task scenarios.

The screenshot shows the Dify Knowledge interface. On the left is a sidebar with 'Documents', 'Pipeline', 'Retrieval Testing', and 'Settings'. The main area is titled 'Documents' and contains a table of knowledge base files.

#	NAME	CHUNKING MODE	WORDS	RETRIEVAL COUNT	UPLOAD TIME	STATUS	ACTION
1	intent-mapping-microros-v2-en.xlsx	GENERAL	553	0	07/29/2026 07:19 PM	Available	[Toggle] [More]
2	decision-layer-training-examples-en.xlsx	GENERAL	2.7k	0	07/29/2026 07:18 PM	Available	[Toggle] [More]

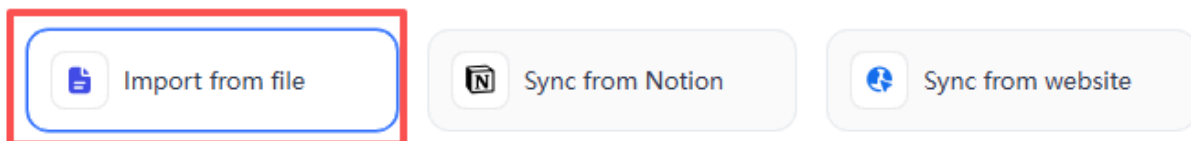
4. Extend the RAG Knowledge Base

- If you need to extend a new knowledge base, click "Create Knowledge".

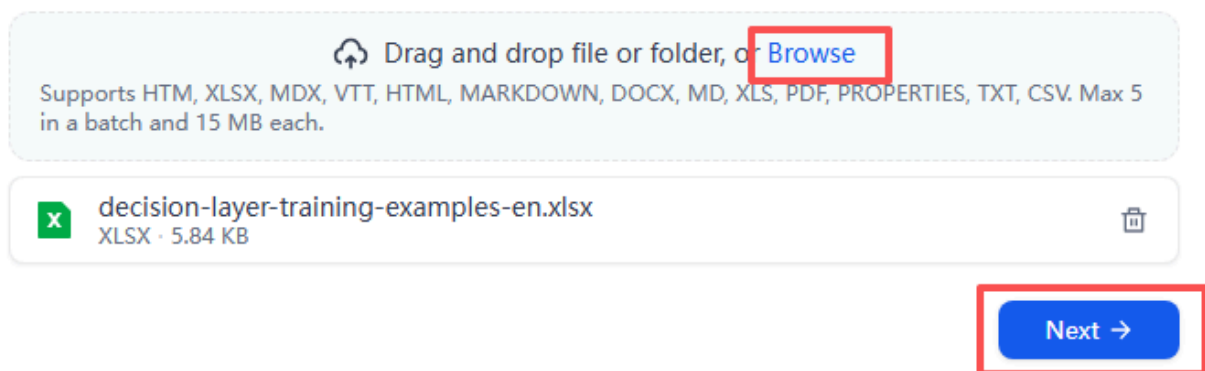


- Here, importing local data is used as the data source example.
- Click "Import from file" —> "Browse" —> "Next".

Data Source



Upload file



[I want to create an empty Knowledge](#)

- Then enter the knowledge base configuration page. Click the preview block to view the chunking effect of the file. Here, the index mode is set to "Economy".

💡 Tip

- For beginners, it is recommended to first use Economy mode for learning and testing. The difference between the two index modes:
 - Economy: retrieves content in the knowledge base through **keywords**, cannot perform

extended retrieval by similar semantics, and the way of retrieving knowledge chunks is relatively rigid.

- High-quality mode: requires an embedding model that consumes extra tokens and requires a rerank model, achieving more accurate similar-semantic chunk retrieval.

4.1 Economy Mode Knowledge Base

- After selecting the following configuration, click "Save & Process".

The screenshot shows the 'KNOWLEDGE' configuration interface. The 'STEP 2 DOCUMENT PROCESSING' tab is active. Under 'Text Preprocessing Rules', the 'Preview Chunk' button is highlighted. In the 'Index Method' section, the 'Economical' option is selected, with a note: 'Using 10 keywords per chunk for retrieval, no tokens are consumed at the expense of reduced retrieval accuracy.' The 'Retrieval Setting' section shows 'Inverted Index' selected and 'Top K' set to 3. On the right, a 'PREVIEW' window displays 24 estimated chunks, with the first few shown as query-answer pairs. At the bottom, the 'Save & Process' button is highlighted.

- Then wait for the embedding to complete and click "Go to Documents".

Knowledge created

We automatically named the Knowledge, you can modify it at any time.

The 'Knowledge name' input field contains the text 'decision-layer-tra...'.

EMBEDDING COMPLETED

The file name 'decision-layer-training-examples-en.xlsx' is displayed with a green checkmark icon on the right.

Chunking Setting	Custom
Maximum Chunk Length	1024
Text Preprocessing Rules	Replace consecutive spaces, newlines and tabs
Index Method	 Economical
Retrieval Setting	 Inverted Index

The 'Access the API' button is on the left, and the 'Go to document →' button is on the right, highlighted with a red border.

- When the knowledge base is normal, the status will show "Available". Then click on the knowledge

base's files.

Documents							
All files of the Knowledge are shown here, and the entire Knowledge can be linked to Dify citations or indexed via the Chat plugin. Learn more!							
All Search		Sort by: Upload Time		Metadata		Add file	
☐	#	NAME	CHUNKING MODE	WORDS	RETRIEVAL COUNT	UPLOAD TIME	STATUS
☐	1	decision-layer-training-examples-en.xlsx	GENERAL	2.7k	0	08/14/2026 12:00 PM	Available

- Then we can see the chunks of the split knowledge base. The small text below each chunk is the automatically split keywords of that chunk (only available in Economy mode).

☐ 24 CHUNKS

☐

Chunk-03 · 240 characters · 0 Retrieval count

query: "Go to points A and B in sequence and see what is there"; answer: "1. Navigate to point A
2. Capture and describe the current view at point A..."
#point #current #view #Go #Navig... #seque... #points #query #Capture #descri...

☐

Chunk-04 · 93 characters · 0 Retrieval count

query: "Return to the starting position"; answer: "1. Navigate back to the starting position"
#starting #back #answer #position #query #Return #Navig...

☐

Chunk-05 · 118 characters · 0 Retrieval count

query: "Where are you now?"; answer: "1. Get the current position
2. Reply with the current position and end the task..."
#answer #task #Reply #position #current #end #Get #query

- If the keywords do not accurately describe the knowledge chunk, click "Edit" on the right side of the chunk to edit the chunk content or keywords individually. The figure below shows modified keywords, and then click "Save".

Edit Chunk

Chunk-03 · 240 characters

query: "Go to points A and B in sequence and see what is there"; answer: "1. N
avigate to point A
2. Capture and describe the current view at point A
3. Navigate to point B
4. Capture and describe the current view at point B
5. End the task"

KEYWORDS

point ×

current ×

view ×

Go ×

Navigate ×

sequence ×

points ×

query ×

Capture ×

describe ×

+ Add keyword

4.2 High-Quality Mode Knowledge Base

- If you need to use a high-quality knowledge base later, refer to the tutorial in this section.
- The knowledge base creation and file import process is the same as before.
- Here, select "High Quality" as the index method, choose any retrieval method, here "Hybrid Search" is used as an example, and finally click "Save & Process".



High Quality RECOMMEND

Calling the embedding model to process documents for more precise retrieval helps LLM generate high-quality answers.



Economical

Using 10 keywords per chunk for retrieval, no tokens are consumed at the expense of reduced retrieval accuracy.

⚠ Once finishing embedding in High Quality mode, reverting to Economical mode is not available.

Embedding Model

multimodal-embedding-v1

Retrieval Setting

[Learn more](#) about retrieval method, you can change this at any time in the Knowledge settings.



Vector Search

Generate query embeddings and search for the text chunk most similar to its vector representation.



Full-Text Search

Index all terms in the document, allowing users to search any term and retrieve relevant text chunk containing those terms.



Hybrid Search RECOMMEND

Execute full-text search and vector searches simultaneously, re-rank to select the best match for the user's query. Users can choose to set weights or configure to a Rerank model.



Weighted Score

By adjusting the weights assigned, this rerank strategy determines whether to prioritize semantic or keyword matching.



Rerank Model

Rerank model will reorder the candidate document list based on the semantic match with user query, improving the results of semantic ranking

qwen3-rerank

⚠ When embedding model supports multi-modal, please select a multi-modal rerank model for better performance.

Top K

3

☐ Score Threshold

0.5

← Previous step

Save & Process

4.3 Recall Testing

- Recall testing tests the actual effect of the knowledge base recalling relevant knowledge chunks based on input, helping tune for better AI model reply effects.
- Open a knowledge base and click "Retrieval Testing" on the left.

decision-layer-tra...

Useful for when you want to answer queries about the decision-layer-training-...

Documents

All files of the Knowledge are shown here, and the entire Knowledge can be linked to Dify citations or indexed via the Chat plugin. [Learn more](#)

All Search Sort by: Upload Time

☐	#	NAME
☐	1	intent-mapping-microros-v2-en.xlsx
☐	2	decision-layer-training-examples-en.xlsx

Documents

Pipeline

Retrieval Testing

Settings

- Enter test content in the source text (simulating the user input content in actual use), then click "Test".
- The recalled chunks and knowledge bases related to the input content will appear on the right. The knowledge base tested here is the Economy mode knowledge base, which retrieves the knowledge base based on keywords.

Tip

The default reference knowledge base at the factory is in High-quality mode.

- For the High-quality mode knowledge base, recalled chunks will have a `SCORE` rating. The higher the score, the better the chunk matches the input content. High-quality mode knowledge bases can perform associative retrieval of similar semantics, but also consume tokens.

Retrieval Test

Test the hitting effect of the Knowledge based on the given query text.

SOURCE TEXT

VECTOR SEARCH

22/200

Track the green blocks

Test

Records

QUERY CONTENT	SOURCE	TIME
Track the green blocks	Retrieval Test	08/14/2026 12:06 PM

3 Retrieved Chunks

Chunk-20

175 characters

SCORE 0.41

query:"Go to point A, find the red ball, and follow it";"answer":1. Navigate to point A

2. Capture the current view...

decision-layer-training-examples-en.xlsx

OPEN

Chunk-18

158 characters

SCORE 0.39

query:"Go to point A, then follow the blue line";"answer":1. Navigate to point A

2. Prepare the camera view for line following...

decision-layer-training-examples-en.xlsx

OPEN

Chunk-19

124 characters

SCORE 0.38

query:"Follow the red ball";"answer":1. Capture the current view

2. Select the red ball's bounding box and start tracking"...

decision-layer-training-examples-en.xlsx

OPEN